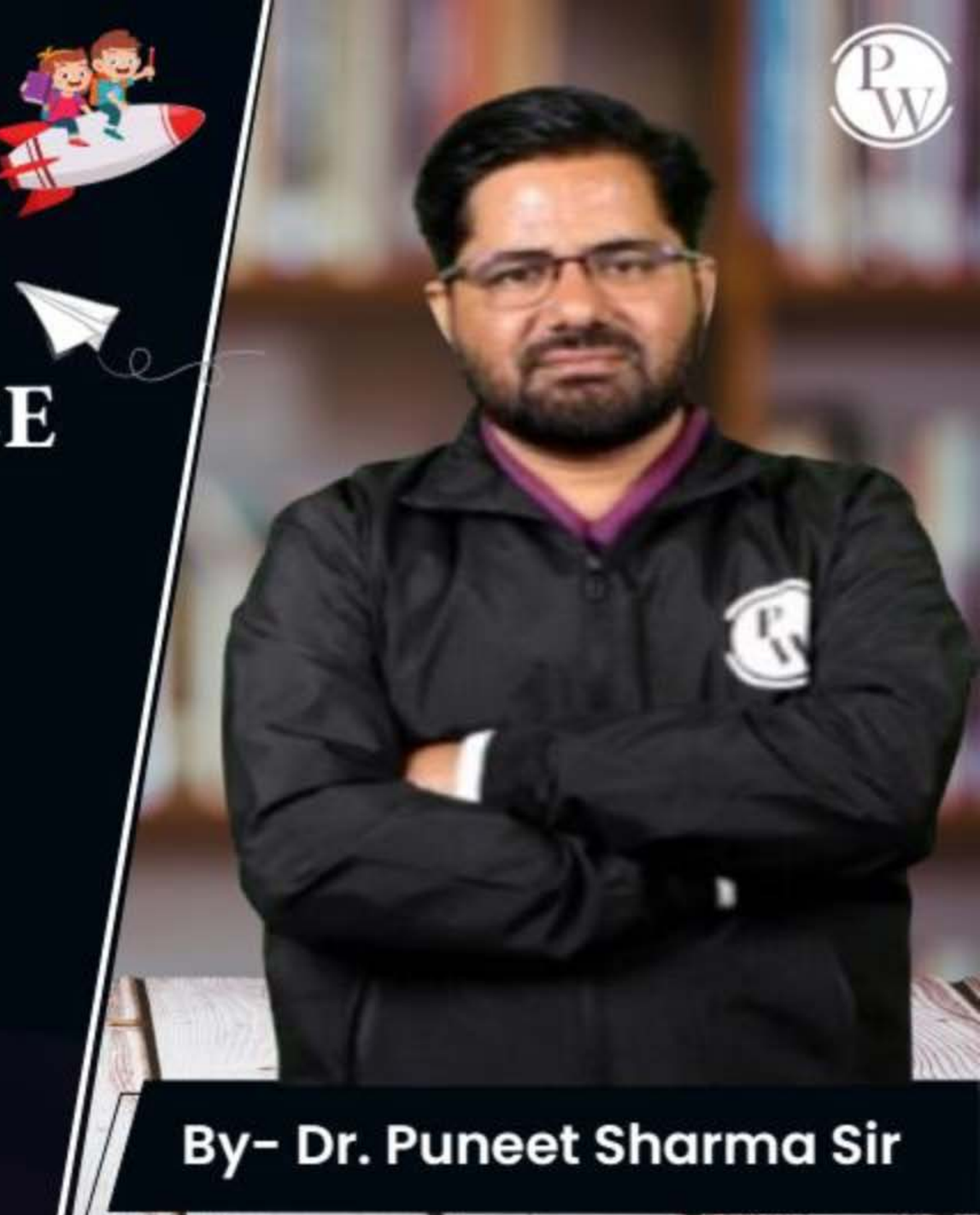
# DATA SCIENCE
# &
# ARTIFICIAL INTELLIGENCE

## Probability and Statistics

## Statistics - 1

Lecture No. 07





By- Dr. Puneet Sharma Sir

# Recap of previous lecture



**Topic**

Continuous Random Variable

(Part 1)

# Topics to be Covered



**Topic**

Continuous Random Variable

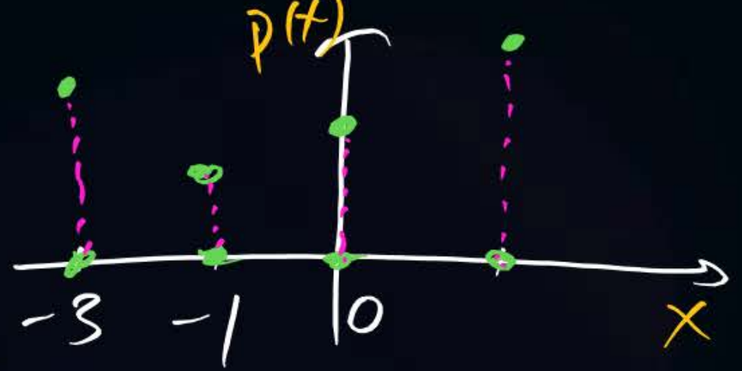
(Part 2)

# Continuous Prob Dist

## D. Prob Dist

Eg

| $X$: | 1 | 0 | −1 | −3 |
|---|---|---|---|---|
| $P(X)$: | $\frac{1}{6}$ | $\frac{5}{36}$ | $\frac{25}{216}$ | $\frac{125}{216}$ |

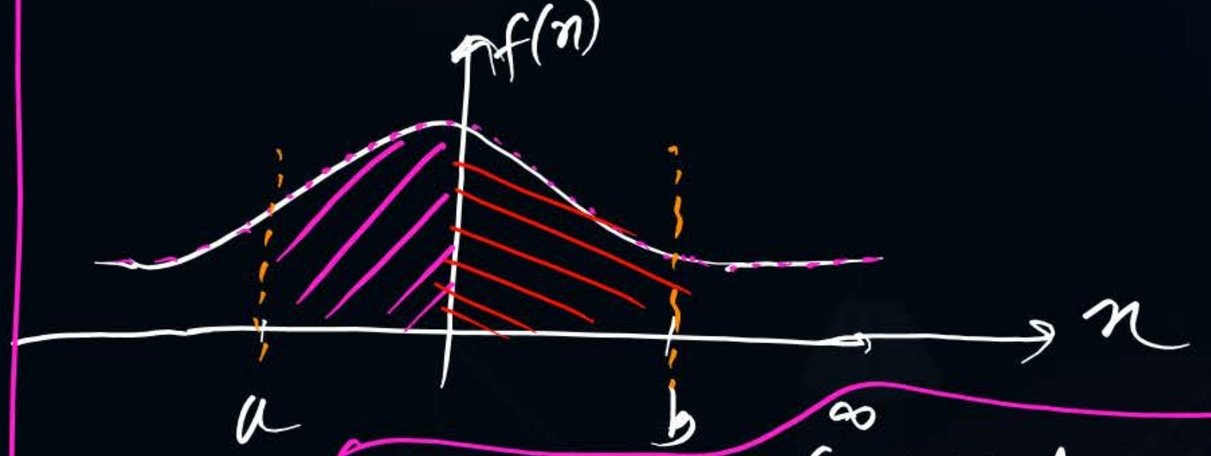



Cond$^{n}$: $p_i \geq 0$ & $\sum p_i = 1$

c.d.f at $(x = -1) = p_1 + p_2 + p_3 = ?$

## C. Prob Dist

Eg Let $-\infty < x < \infty$ Then $f(x)$ = p.d.f for $x$




Cond$^{n}$: $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x)\,dx = 1$

c.d.f at $(x = a) = F(a) = \int_{-\infty}^{a} f(x)\,dx$

" at $(x = b) = F(b) = \int_{-\infty}^{b} f(x)\,dx$

**Note** if $x$ is C.R.V s.t it's p.d.f is $f(x)$ then we have following results;




(1) $f(x) \cong$ Prob at $x = ??$

(2) $P(-\infty < x < \infty) = 1$

or $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\Rightarrow$ Total area under $f(x)$ will be of 1 units

(3) $P(a \le x \le b) = \int_{x=a}^{b} f(x)\,dx$ = Area under $f(x)$ b/n $x=a$ & $x=b$

(4) $f(x) \cong$ Prob at $x$ = (Area from $-\infty$ to $x$) − (Area from $-\infty$ to $\bar{x}$)

$= F(x) - F(\bar{x})$

## Remaining Theory of C.R.V ⟹

① $\text{Mean}(x) = E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$

Similarly $E(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = 2^{nd}$ Moment

$E(x^3) = \int_{-\infty}^{\infty} x^3 f(x)\,dx = 3^{rd}$ Moment

$E(|x|) = \int_{-\infty}^{\infty} |x| f(x)\,dx$ .....

Let $f(x)$ is p.d.f of $x$ and $g(x)$ be another function of $x$

Then

$$E\{g(x)\} = \int_{-\infty}^{\infty} g(x) \cdot f(x)\,dx$$

② $\boxed{\text{Var}(x) = E(x^2) - E^2(x)}$

③ $SD(\sigma) = \pm\sqrt{\text{Var}(x)}$

# Cummulative Density fun$^n$ (C.d.f)

Let $x$ is C.R.V, $f(x)$ is it's p.d.f

then it's c.d.f is denoted by $F(x)$ and is defined as;

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$F(-\infty) = \int_{-\infty}^{-\infty} f(x)\,dx = 0$$

$$F(\infty) = \int_{-\infty}^{\infty} f(x)\,dx = 1$$

i.e $0 \le F(x) \le 1$

eg $F(a) = \int_{-\infty}^{a} f(x)\,dx$ = c.d.f at $(x=a)$

⊛ Graph of p.d.f should not lies below X axis & $f(x)$ may be $> 1$ also

while " of C.D.F " " " " X axis But $0 \le F(x) \le 1$

(*) Relation b/w p.d.f & c.d.f $\Rightarrow$

p.d.f $f(x)$ , c.d.f $F(x)$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$f(x) = \frac{d}{dx} F(x)$$

**Qe** if $x$ is C.R.V s.t it's p.d.f is $f(x) = ke^{-\alpha|x|}$ ; $\alpha \in R^+$ then find $k = ?$

**Sol:** w.k that area under $f(x) = 1$

i.e $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\int_{-\infty}^{\infty} ke^{-\alpha|x|}\,dx = 1$ (Even func$^n$)

$2\int_{0}^{\infty} ke^{-\alpha|x|}\,dx = 1$

$2k\int_{0}^{\infty} e^{\alpha(+x)}\,dx = 1$

$2k\int_{0}^{\infty} e^{-\alpha x}\,dx = 1$

$2k\left[\frac{e^{-\alpha x}}{-\alpha}\right]_0^{\infty} = 1$

$-\frac{2k}{\alpha}\left[e^{-\infty} - e^{0}\right] = 1$

$-\frac{2k}{\alpha}\left[0 - 1\right) = 1 \Rightarrow k = \frac{\alpha}{2}$ Ans

Q: If $f(x) = e^{-2|x|}$ is Valid p.d.f for C.R.V $x$ then $P(-1 \le x \le 1) = ?$

Sol: $P(-1 \le x \le 1) = \int_{-1}^{1} f(x)\,dx$

$= \int_{-1}^{1} e^{-2|x|}\,dx = 2\int_{0}^{1} e^{-2|x|}\,dx$

($e^{-2|x|}$ : Even func$^n$)

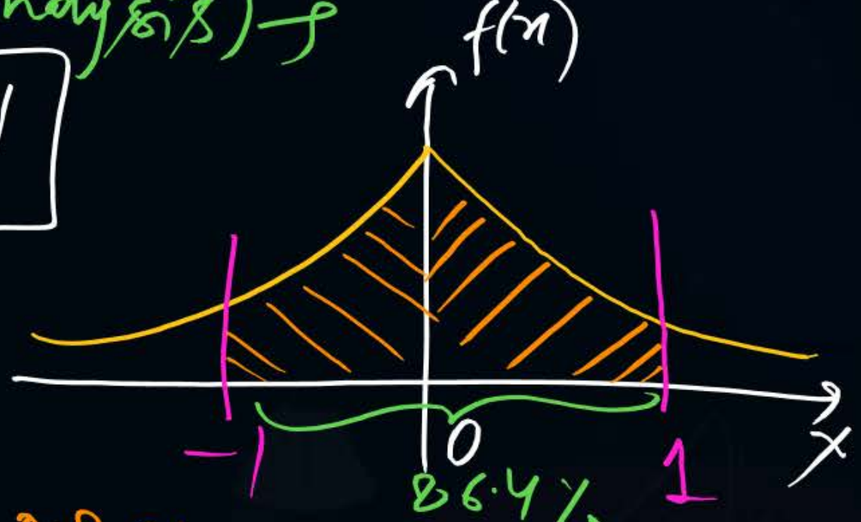
$= 2\int_{0}^{1} e^{-2(+x)}\,dx = 2\int_{0}^{1} e^{-2x}\,dx$

$= 2\left[\frac{e^{-2x}}{-2}\right]_0^1 = -\left[e^{-2} - e^{0}\right]$

$= 1 - \frac{1}{e^2} = 0.864$

PODCAST (Analysis) →

$f(x) = e^{-2|x|}$




$\because P(-1 \le x \le 1) = 0.864$

Area under $f(x)$ b/w $-1$ & $1$ $= 0.864$

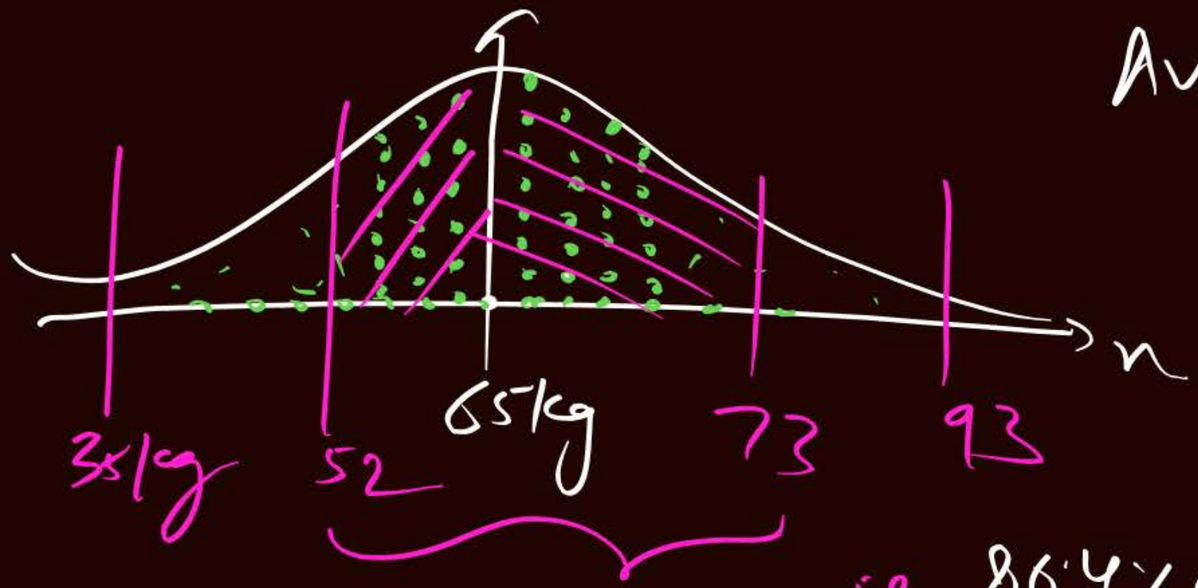
i.e 86.4% area under $f(x)$ lies in b/w $-1$ & $1$

$\because P(-1 \le x \le 1) = \frac{0.864}{1} = \frac{86.4}{100} \approx 86.4\%$

PODCAST (ANALYSIS) →

let x = weight : 63 65 63 64 72, 52, ....., 89, 43, 61

500 students




Av = 65kg

Total Person = 500 ≃ 100% = 1

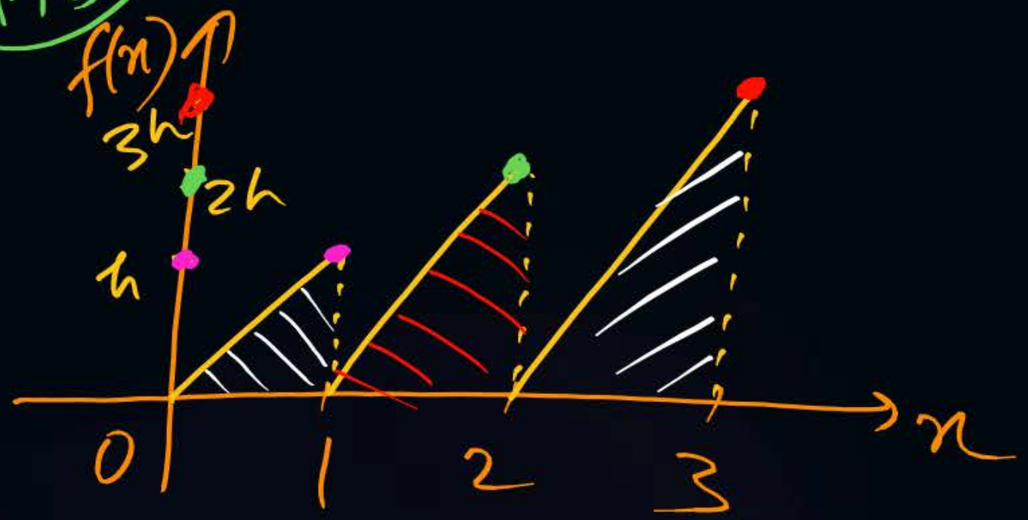
let P(52 < n < 73) ??? = 0.864

= 86.4% area

i.e 86.4% area ≃ 86.4% × 500P = 432 persons.

Conclusion: 432 persons weight lies in b/n 52kg & 73kg

Q. (IIT-B) If $f(x)$ is density func$^n$ ($= p.d.f$) of $x$ shown in the diagram then find $h = ?$




W.k. that, Total area under $f(x) = 1$

$$\text{Area}\left[\Delta_1 + \Delta_2 + \Delta_3\right] = 1$$

$$\frac{1}{2}\left[1 \times h + 1 \times 2h + 1 \times 3h\right] = 1$$

$$3h = 1 \Rightarrow \boxed{h = \frac{1}{3}}$$

HW8 If $f(x)$ is p.d.f for R.V '$x$' then $k=$ ? where $f(x)=\begin{cases} kx+1, & 0<x<4 \\ 0, & \text{otherwise} \end{cases}$

(a) $-1/2$

(b) $-3/8$

(c) $-3/16$

(d) $f(x)$ can not be p.d.f

$\because f(x)$ is p.d.f then

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^4 (kx+1)\,dx = 1$$

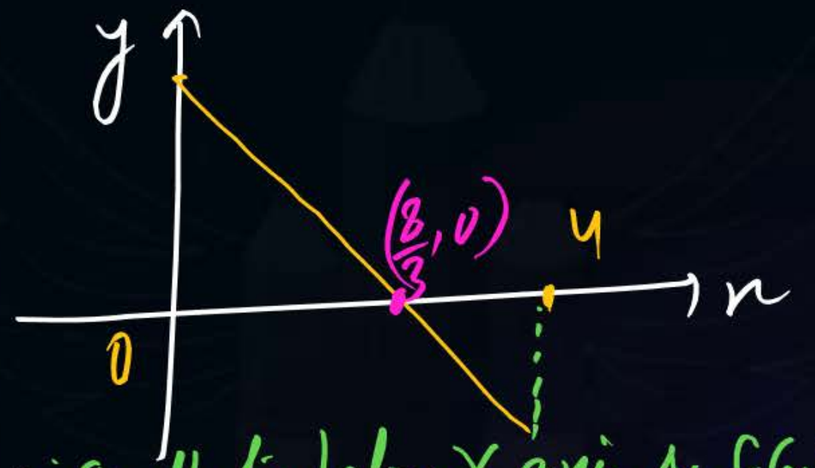
$$\left[k\left(\frac{x^2}{2}\right)+x\right]_0^4 = 1$$

$$\left[\frac{k}{2}(16-0)+(4-0)\right]=1$$

$$k=-3/8$$

Let us cross check it

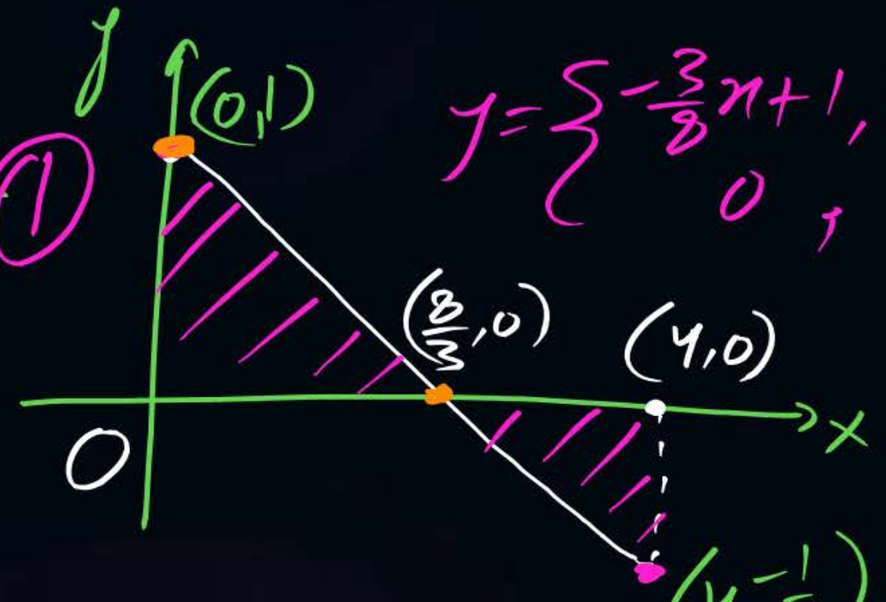
$$y=f(x)=\begin{cases} \frac{-3}{8}x+1, & 0<x<4 \\ 0, & \text{otw} \end{cases}$$




$\therefore$ graph lies below X axis so $f(x)$ can not be a p.d.f for $x$

Analysis :-

(1)




$$y=\begin{cases}-\frac{3}{8}x+1, & 0\le x\le 4\\ 0, & \text{otw}\end{cases}$$

Total area $=\frac{1}{2}\times\frac{8}{3}\times(1)+\frac{1}{2}\times\left(4-\frac{8}{3}\right)\times\frac{1}{2}$

$=\frac{8}{6}+\frac{2}{6}=\frac{10}{6}=\frac{5}{3}>1$

i.e. $f(x)$ can not be a p.d.f

(3) $y=-\frac{3}{8}x+1\Rightarrow\frac{3}{8}x+y=1\Rightarrow\boxed{\frac{x}{8/3}+\frac{y}{1}=1}$

Analysis 2 :-

while in case of Integration

$$\int_0^4 f(x)\,dx=\int_0^4\left(-\frac{3}{8}x+1\right)dx$$

$$=\int_0^{8/3}\left(-\frac{3}{8}x+1\right)dx+\int_{8/3}^{4}\left(-\frac{3}{8}x+1\right)dx$$

$$=\frac{8}{6}+\left(-\frac{2}{6}\right)^{*}=1$$

these things are only to develop Neurons of Brain.

Q. If $f(x)=\begin{cases}1+x, & -1\le x\le 0\\ 1-x, & 0\le x\le 1\end{cases}$ then find Graph of $f(x)$

Sol: Case I: $-1\le x\le 0$

$y=1+x$

$-x+y=1$

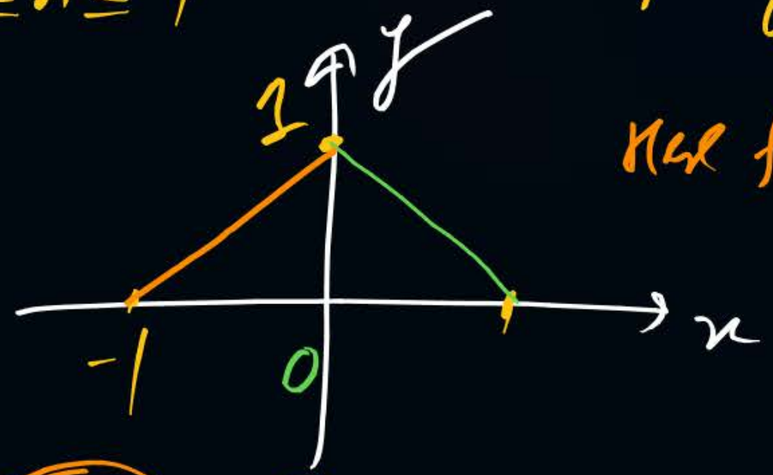
$\frac{x}{-1}+\frac{y}{1}=1$

Case II: $0\le x\le 1$

$y=1-x$

$x+y=1$

$\frac{x}{1}+\frac{y}{1}=1$



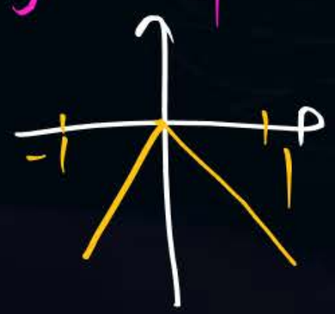

Here $f(x)$ is an Even func$^n$.

M-II $y=\begin{cases}1+x, & -1\le x\le 0\\ 1-x, & 0\le x\le 1\end{cases}=1-|x|\ ;\ -1\le x\le 1$

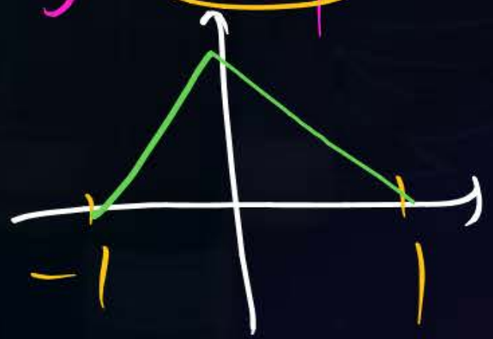
$y=|x|$




$y=-|x|$




$y=-|x|+1\ ;\ -1\le x\le 1$



Q. If $f(x)=\begin{cases}1+x, & -1\le x\le 0\\ 1-x, & 0\le x\le 1\end{cases}$ is density func$^n$ for R.V 'x' then evaluate

Mean(x), 2nd Moment, Var(x), S.D(σ) ?

Sol: (1) $\text{Mean}(x)=E(x)=\int_{-\infty}^{\infty} x f(x)\,dx=\int_{-1}^{1} \underbrace{x.f(x)}_{\text{odd func}^n}\,dx=0$

(2) $2^{nd}\text{ Moment}=E(x^2)=\int_{-\infty}^{\infty} x^2 f(x)\,dx=\int_{-1}^{1} \underbrace{x^2 f(x)}_{\text{Even func}^n}\,dx=2\int_0^1 x^2 f(x)\,dx=2\int_0^1 x^2(1-x)\,dx$

$=2\int_0^1 (x^2-x^3)\,dx=2\left[\frac{x^3}{3}-\frac{x^4}{4}\right]_0^1=2\left[\frac{1}{3}-\frac{1}{4}\right]=\frac{1}{6}$

(3) $\text{Var}(x)=E(x^2)-E^2(x)=\frac{1}{6}-(0)^2=\frac{1}{6}$

(4) $S.D(\sigma)=\pm\frac{1}{\sqrt{6}}$

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Q: If $P_X(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ is p.d.f for $x$ & $g_x(x) = e^{\frac{3x}{4}}$ is another func$^n$ of $x$

then find $E\{g_x(x)\} = ?$

Sol: Let $P_X(x) = f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otw} \end{cases}$ & let $g_x(x) = g(x) = e^{\frac{3x}{4}}$

Now $E\{g(x)\} = \int_{-\infty}^{\infty} g(x) f(x)\,dx = \int_{-\infty}^{0} g(x)(0)\,dx + \int_{0}^{\infty} g(x) \cdot (e^{-x})\,dx = \int_{0}^{\infty} e^{\frac{3x}{4}} \cdot e^{-x}\,dx$

$$= \int_{0}^{\infty} e^{-\frac{x}{4}}\,dx = \left( \frac{e^{-x/4}}{-1/4} \right)_0^{\infty} = -4\left[ e^{-\infty} - e^{0} \right] = -4(0-1) = 4$$

Eg: if $f(x)=e^{-2|x|}$ is p.d.f for C.R.V $x$, then evaluate ① $F(-3)$, ② $F(0)$ ③ $F(5)=?$

Sol: W.k.t that $\boxed{F(x)=\int_{-\infty}^{x} f(x)\,dx}$

① $F(-3)=\int_{-\infty}^{-3} e^{-2|x|}\,dx=\int_{-\infty}^{-3} e^{-2(-x)}\,dx=\int_{-\infty}^{-3} e^{2x}\,dx=\left(\frac{e^{2x}}{2}\right)_{-\infty}^{-3}=\frac{1}{2}\left(e^{-6}-0\right)=\frac{1}{2e^6}$

② $F(0)=\int_{-\infty}^{0} e^{-2|x|}\,dx=\int_{-\infty}^{0} e^{-2(-x)}\,dx=\int_{-\infty}^{0} e^{2x}\,dx=\left(\frac{e^{2x}}{2}\right)_{-\infty}^{0}=\frac{1}{2}(1-0)=\frac{1}{2}$

③ $F(5)=\int_{-\infty}^{5} e^{-2|x|}\,dx=\int_{-\infty}^{0} e^{-2|x|}\,dx+\int_{0}^{5} e^{-2|x|}\,dx=\frac{1}{2}+\int_{0}^{5} e^{-2(+x)}\,dx=\frac{1}{2}+\left(\frac{e^{-2x}}{-2}\right)_{0}^{5}$

$=\frac{1}{2}-\frac{1}{2}\left(e^{-10}-1\right)=1-\frac{1}{2e^{10}}$

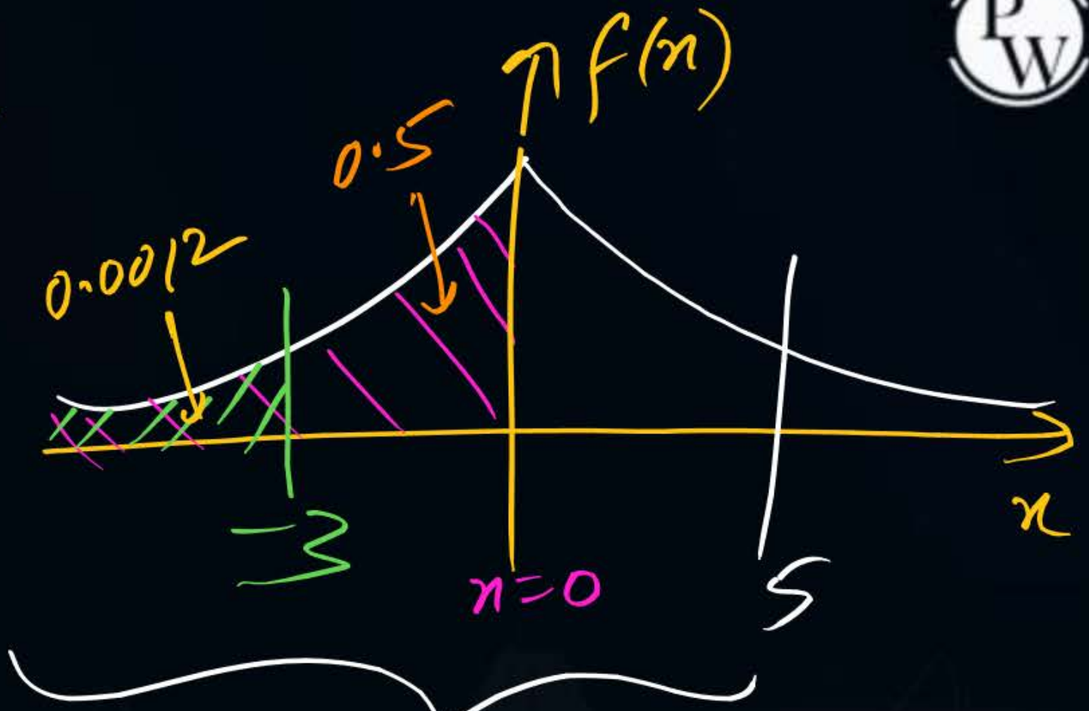
Analysis:

① $f(-3) = \frac{1}{2e^6} = 0.0012$

② $f(0) = \frac{1}{2} = 0.5 = 50\%$

③ $f(5) = 1 - \frac{1}{2e^{10}} = 0.999$

p.d.f $\boxed{f(x) = e^{-2|x|}}$




Prob $= f(5) = 0.999 = 99.9\%$

Q: If $f(x)=e^{-2|x|}$ is p.d.f for R.V x then find it's C.d.f for any $x>0$ ?

Sol: for $x>0$:

$$F(x)=\int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{x} e^{-2|x|}\,dx = \int_{-\infty}^{0} e^{-2|x|}\,dx + \int_{0}^{x} e^{-2|x|}\,dx$$

$$= \int_{-\infty}^{0} e^{-2(-x)}\,dx + \int_{0}^{x} e^{-2(+x)}\,dx$$

$$= \int_{-\infty}^{0} e^{2x}\,dx + \int_{0}^{x} e^{-2x}\,dx$$

$$= \left(\frac{e^{2x}}{2}\right)_{-\infty}^{0} + \left(\frac{e^{-2x}}{-2}\right)_{0}^{x}$$

$$= \frac{1}{2}(1-0) - \frac{1}{2}\left(e^{-2x}-1\right)$$

$$F(x)=1-\frac{1}{2e^{2x}}$$

Cross check: $F(5)=1-\frac{1}{2e^{10}}$

if $f(x) = e^{-2|x|}$ then find C.d.f for $x$ if ① $x < 0$ ② $x = 0$ ③ $x > 0$

Soln: ① for $x < 0$:

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{x} e^{-2|x|}\,dx$$

$$f(x) = \int_{-\infty}^{x} e^{-2(-x)}\,dx = \int_{-\infty}^{x} e^{2x}\,dx = \left(\frac{e^{2x}}{2}\right)_{-\infty}^{x}$$

$$= \frac{1}{2}\left(e^{2x} - e^{-\infty}\right) = \frac{1}{2}e^{2x}$$

Cross check: $F(-3) = \frac{1}{2}e^{-6} = \frac{1}{2e^6}$

② At $x = 0$:

$$F(0) = \int_{-\infty}^{0} f(x)\,dx = \int_{-\infty}^{0} e^{-2|x|}\,dx$$

$$= \int_{-\infty}^{0} e^{+2x}\,dx = \left(\frac{e^{2x}}{2}\right)_{-\infty}^{0} = \frac{1}{2}$$

**Note:**

① if $f(x) = e^{-2|x|}$ is p.d.f then $F(x) = \begin{cases} \frac{e^{2x}}{2} & , x < 0 \\ \frac{1}{2} & , x = 0 \\ 1 - \frac{1}{2e^{2x}} & , x > 0 \end{cases}$

② if $f(x) = \begin{cases} \frac{1}{2}e^{2x} & , x < 0 \\ \frac{1}{2} & , x = 0 \\ 1 - \frac{1}{2e^{2x}} & , x > 0 \end{cases}$ then find p.d.f of $x$ is $f(x) = ?$

Sol: $f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{2e^{2x}}{2} & , x < 0 \\ -\frac{e^{2x}(-2)}{2} & , x > 0 \end{cases} = \begin{cases} e^{2x} & , x < 0 \\ e^{-2x} & , x > 0 \end{cases} = e^{-2|x|}$

# THANK - YOU